HEART DISEASE PREDICTION USING MACHINE LEARNING

MINI PROJECT REPORT

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ABSTRACT

Heart disease is one of the leading causes of death worldwide. Early detection and diagnosis can significantly reduce mortality rates and improve patient outcomes. This project presents a Machine Learning based system for predicting heart disease using patient health parameters. Various medical attributes such as age, cholesterol level, blood pressure, heart rate, and chest pain type are analyzed to determine the likelihood of heart disease. Machine Learning algorithms are trained on historical medical data to provide accurate predictions. The proposed system assists healthcare professionals in making informed decisions and enables early preventive measures.

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1. INTRODUCTION

Heart disease refers to various conditions that affect the heart and blood vessels. It remains one of the major health concerns globally. Traditional diagnosis methods often require extensive medical examinations and expert analysis.

Machine Learning offers an intelligent approach for analyzing healthcare data and identifying hidden patterns. By utilizing historical patient records, predictive models can be developed to classify whether a patient is at risk of heart disease.

This project aims to build a heart disease prediction system using Machine Learning techniques that provide accurate and reliable predictions.

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2. PROBLEM STATEMENT

Diagnosing heart disease at an early stage is challenging due to the complexity of medical data and various influencing factors. There is a need for an automated prediction system that can analyze patient information efficiently and assist doctors in diagnosis.

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3. OBJECTIVES

- Predict heart disease using Machine Learning algorithms.
- Analyze patient medical data.
- Improve prediction accuracy.
- Assist healthcare professionals in decision-making.
- Reduce diagnosis time and effort.

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4. LITERATURE SURVEY

Several researchers have applied Machine Learning techniques in healthcare for disease prediction. Algorithms such as Logistic Regression, Decision Trees, Random Forests, and Support Vector Machines have shown promising results in predicting heart disease.

Studies indicate that Machine Learning can improve prediction accuracy and help identify patients at risk before severe complications occur.

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5. SYSTEM REQUIREMENTS

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB or above
- Storage: 20 GB Free Space

Software Requirements

- Python 3.x
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

6. DATASET DESCRIPTION

The dataset contains patient medical information used for prediction.

Attributes

Attribute	Description
Age	Age of patient
Sex	Gender
CP	Chest pain type
Trestbps	Resting blood pressure
Chol	Cholesterol level
FBS	Fasting blood sugar
RestECG	ECG results
Thalach	Maximum heart rate
Exang	Exercise-induced angina
Oldpeak	ST depression
Target	Heart disease status

7. METHODOLOGY

The project follows the following steps:

Step 1: Data Collection

Collect heart disease dataset containing patient health records.

Step 2: Data Preprocessing

- Handle missing values
- Remove duplicate records
- Normalize data

Step 3: Feature Selection

Select important features affecting heart disease.

Step 4: Model Training

Train Machine Learning algorithms using training data.

Step 5: Model Evaluation

Evaluate model performance using testing data.

Step 6: Prediction

Predict whether a patient is likely to have heart disease.

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8. ALGORITHMS USED

Logistic Regression

A statistical classification algorithm used for binary outcomes.

Decision Tree

A tree-based algorithm that makes decisions using feature conditions.

Random Forest

An ensemble learning technique that combines multiple decision trees to improve accuracy.

K-Nearest Neighbors (KNN)

Classifies data based on similarity to nearby data points.

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9. SYSTEM ARCHITECTURE

Patient Data

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Data Preprocessing

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Feature Selection

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Machine Learning Model Training

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Model Evaluation

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Heart Disease Prediction

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10. IMPLEMENTATION

The system is implemented using Python and Machine Learning libraries.

Libraries Used

- Pandas
- NumPy
- Scikit-learn
- Matplotlib

The dataset is divided into training and testing sets. The selected algorithm is trained and evaluated using accuracy metrics.

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11. RESULTS

The Machine Learning model successfully predicts heart disease based on patient medical records.

Sample Output

Prediction Result:

- Heart Disease Detected OR

- No Heart Disease

The Random Forest algorithm achieved high prediction accuracy and reliable performance.

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12. ADVANTAGES

- Early disease detection
- Faster diagnosis
- High prediction accuracy
- Supports medical professionals
- Cost-effective solution

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13. LIMITATIONS

- Accuracy depends on dataset quality.
- Requires sufficient training data.
- Cannot completely replace medical experts.

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14. FUTURE SCOPE

- Real-time patient monitoring.
- Mobile healthcare applications.
- Integration with hospital management systems.
- Deep Learning based prediction models.
- Cloud-based healthcare analytics.

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15. CONCLUSION

Heart Disease Prediction using Machine Learning provides an effective solution for early diagnosis and risk assessment. By analyzing patient health parameters, the system predicts the likelihood of heart disease with good accuracy. This technology can assist healthcare professionals in making faster and more informed decisions, ultimately improving patient care and reducing health risks.

REFERENCES

1. Heart Disease UCI Dataset.
2. Python Documentation.
3. Scikit-learn Documentation.
4. Research Papers on Machine Learning in Healthcare.
5. Machine Learning Books and Journals.